ASRC Fuel Tank Robot Daily Report

Closed-loop control: operating pretension, refined path and an exact 20 degree reference, 12 November 2026

This morning the team completed the first closed-loop round trip (v1): head-angle and rope-tension feedback with nominal-model gains, passing all settled checks in five cases, and reorganised the repository. Two items were left open: rope tension during command transitions (down to 0.57 N, with the controller aiming at only 1.5 N), and node movement above the 5 mm sampling bound. Today's work closed both, then removed a reference error that caused most of the remaining angle error during motion.

Summary of today's work

- **Project renewed after the reorganisation.** Links (45 Markdown files) and model tests checked; the new M end-disk coupon moved next to the STL it loads (cad/previews/u4m2/); model data paths refreshed. No other problems found.
- **Closed loop v2: operating pretension and refined path.** Same model, reference, gains and cases. The correction keeps every rope at 3 N or more; tensions are predicted before each new command and a small pre-correction is added if any would fall below 3 N; every move is subdivided until no point moves more than 4.5 mm, and every intermediate state is solved and checked. Lowest rope tension in any recorded state: 2.62 N (v1: 0.57 N); largest node move 4.5 mm (v1: 6.8 mm).
- **Closed loop v3: exact 20 degree reference.** The 0 and 5 degree yaw poses of the 3 N reference had been accepted at 16 to 17 degrees. Regenerated at exactly 20 degrees; the reference cycle's largest elevation error fell from 3.93 to 0.18 degrees. **Four of five cases now pass every check in every recorded state.**
- **Correction during moves** was tried for the 550 g head at large yaw and made it worse; it is not recommended.

Case	v1: lowest tension / largest move	v2	v3: states > 1 deg in motion	v3: all checks, every state
Nominal	0.93 N / 5.2 mm	2.73 N / 2.7 mm	0 (max 0.18 deg)	**Pass**
Rod E 75 GPa	0.57 N / 5.5 mm	2.62 N / 2.5 mm	0 (max 0.61 deg)	**Pass**
Head 550 g	0.84 N / 6.8 mm	2.76 N / 4.5 mm	202 (max 3.79 deg)	Angle in motion only
Free rope 1.5 m	0.91 N / 5.1 mm	2.72 N / 2.5 mm	0 (max 0.25 deg)	**Pass**
Rope EA -20%	0.96 N / 5.2 mm	2.73 N / 2.6 mm	0 (max 0.28 deg)	**Pass**

Decisions from today
Use the v2 control sequence (3 N correction target, tension-aware arrival, refined path) with the exact 20 degree reference (v3). Do not correct during moves. The remaining 550 g large-yaw angle error needs gains that adapt to measured responses, not more frequent correction.

Closed loop v2: operating pretension and refined path

The controller still observes only head yaw, elevation and the 12 rope tensions; plant parameters are not used. Changes to the control sequence of v1:

- **Operating tension.** The correction keeps every predicted rope tension at 3 N or more (v1: 1.5 N). A sample is settled at an angle error of at most 0.2 degrees and a measured minimum tension of at least 2.5 N.
- **Tension-aware arrival.** Before each new command, tensions are predicted from the measured tensions and the nominal gains; if any would fall below 3 N, a pre-correction of at most 0.5 mm per rope is added that disturbs the head angles as little as possible.
- **Refined path.** Every move is applied in rope steps of at most 0.25 mm, halved until no centreline node moves more than 4.5 mm. Every intermediate state is solved from the previous one and checked (stability, tension, geometry, strain, 3D tank clearance), and checked against the target interpolated along the move.

Case	v1 lowest tension	v1 states below 2.5 N	v2 lowest tension	v2 states below 2.5 N	Largest node move v1 / v2	v2 peak tension
Nominal	0.93 N	1577 of 1709	**2.73 N**	**0** of 3604	5.25 / **2.74** mm	27.1 N
Rod E 75 GPa	0.57 N	1321 of 1941	**2.62 N**	**0** of 3840	5.45 / **2.50** mm	27.3 N
Head 550 g	0.84 N	1896 of 2465	**2.76 N**	**0** of 4093	6.82 / **4.49** mm	27.1 N
Free rope 1.5 m	0.91 N	1493 of 1775	**2.72 N**	**0** of 3628	5.09 / **2.50** mm	25.1 N
Rope EA -20%	0.96 N	1493 of 1772	**2.73 N**	**0** of 3627	5.20 / **2.62** mm	26.3 N

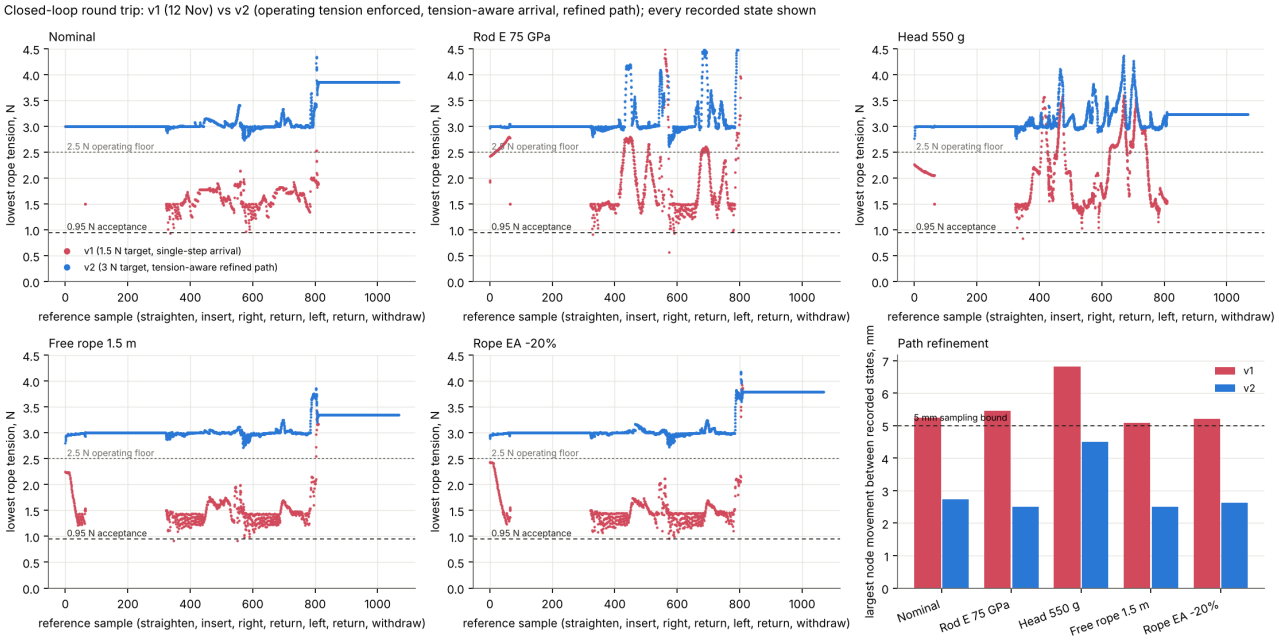


Figure 1 Lowest rope tension of every recorded state, v1 (red) and v2 (blue), and the largest node movement between recorded states.

- Every recorded state of every case stays above the 2.5 N operating floor; no state is unstable or fails geometry, strain or tank clearance; all samples settle within 0.2 degrees (v1 550 g: 0.70 degrees). Cost: peak rope tension 25 to 27 N.
- Remaining in v2: temporary angle error during motion, at the straightening-to-insertion hand-off in every case (up to 3.3 degrees) and at 155 to 175 degrees yaw with the 550 g head (up to 2.85 degrees).

Closed loop v3: exact 20 degree reference

The hand-off error came from the reference itself: the 0 and 5 degree poses of the 3 N yaw sweep had been accepted on the fine mesh at 16.9 and 16.1 degrees, because acceptance allowed plus or minus 5 degrees. The sweep was regenerated requiring the fine-mesh elevation within 1 degree of 20 (all 36 poses now at 20.0 degrees), and the reference cycle and nominal gains were rebuilt from it.

Case	States > 1 deg in motion, v2 / v3	Largest angle in motion, v2 / v3	Lowest tension v3	Peak tension v2 / v3	Correction used v2 / v3
Nominal	5 / **0**	3.08 / **0.18** deg	2.98 N	27.1 / 17.7 N	2.89 / 0.05 mm
Rod E 75 GPa	6 / **0**	3.18 / **0.61** deg	2.92 N	27.3 / 22.5 N	3.39 / 3.00 mm
Head 550 g	87 / **202**	3.01 / **3.79** deg	2.76 N	27.1 / 23.7 N	3.74 / 3.26 mm
Free rope 1.5 m	6 / **0**	3.26 / **0.25** deg	2.81 N	25.1 / 18.0 N	3.62 / 0.90 mm
Rope EA -20%	6 / **0**	3.10 / **0.28** deg	2.89 N	26.3 / 17.9 N	3.66 / 0.72 mm

v3 results shown with correction during moves enabled (0.5 degree threshold, kept only if the measured error decreases); it was used only in the 75 GPa and 550 g cases.

Head 550 g: four versions

Head 550 g	States > 1 deg in motion	Largest
v2 (old reference)	87	3.01 deg
v3, fixed reference only	115	3.13 deg
v3, fixed reference + guarded correction during moves	202	3.79 deg
v3, unguarded first attempt (diagnostic)	872	6.82 deg

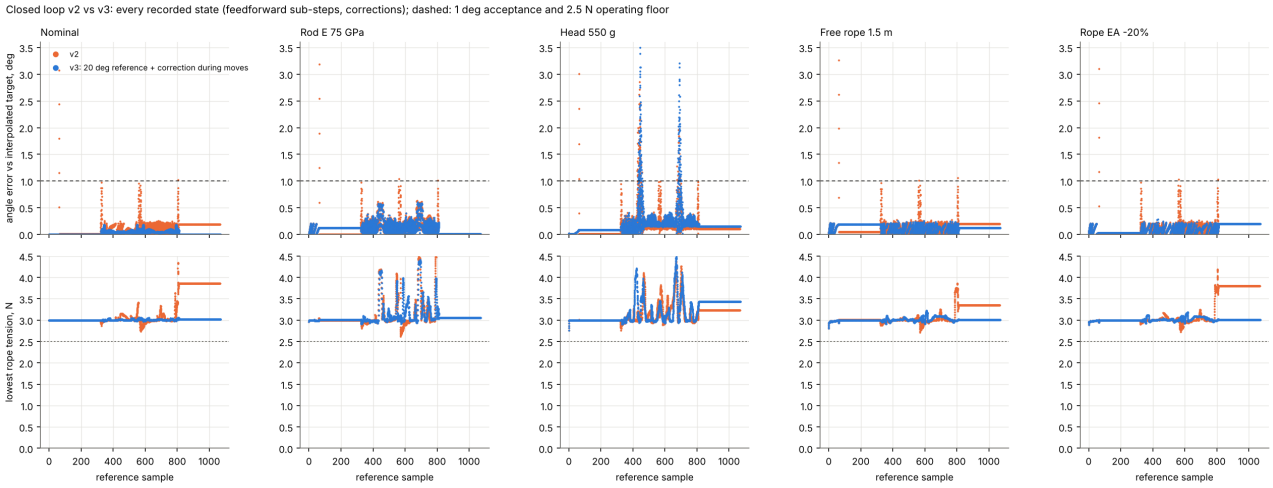


Figure 2 Angle error against the interpolated target and lowest rope tension for every recorded state, v2 (orange) and v3 (blue).

- **Nominal, 75 GPa rod, 1.5 m free rope and rope EA -20% pass every check in every recorded state**: angle error during motion at most 0.61 degrees, rope tension at least 2.81 N, node move at most 2.5 mm. Peak tension falls to 17.7 to 22.5 N because the controller no longer fights a wrong reference.
- **Head 550 g:** temporary angle error remains at 155 to 175 degrees yaw. The nominal gains have the wrong sign there for the heavier head: a correction for 0.2 degrees moves the head about 1 degree the other way, so correcting more often adds overshoot. These states pass every tension, stability, geometry and clearance check, and all samples settle within 0.2 degrees.

Limitations

- Quasi-static model with ideal measurements: no motor dynamics, sampling time, sensor noise or delay, friction, backlash or NiTi hysteresis. Positive static stiffness does not establish dynamic closed-loop stability.
- The five cases were used while developing the controller changes; they are not an independent validation set.
- Rod properties are calculated, not measured; the head is a 500 g point mass with a 63 mm envelope; tank clearance is sampled at recorded states.

Next steps

Priority	Task	Purpose
1	Online gain update	Update the controller gains from measured head-angle responses (for example a Broyden update) to remove the 550 g large-yaw error
2	Dynamics and sensing	Winch speed and travel limits, control update rate, sensor noise and delay; repeat the five cases
3	Build and test the backbone M module	Continuous 96 mm module, full-length rod: stiffness, buckling, end-joint slip (test plan, end-disk coupon)
4	Rope and winch calibration	Measured rope stiffness, free length and friction
5	Head CAD, entry, storage, rear mechanism	As in the roadmap

Scripts and data: simulations/u4m2/backbone_20261108 (README sections 12 and 13; closed_loop_v2.py, summarize_closed_loop_v2.py, test_closed_loop_v2.py, min_force_sweep.py --elev-tol; results in closed_loop_v2_results, closed_loop_v3_results and length_cycle_results_pre3_tol1). This report is generated by reports/builders/build_report_2026_11_12.py.